GATE-ALL BRANCHES



ENGINEERING MATHEMATICS

Probability and Statistics

Lecture No.- 09



By- Rahul sir

Topics covered in previous lecture



Topic Problems based on Bayes theorem

Topic Concept of random variable

Topic Types of random variable

Topics to be Covered



Topic Concept of Random variable

Topic Problems based on random variable

10th

Random variable:

$f : X \longrightarrow Y$ (mathematical Function)

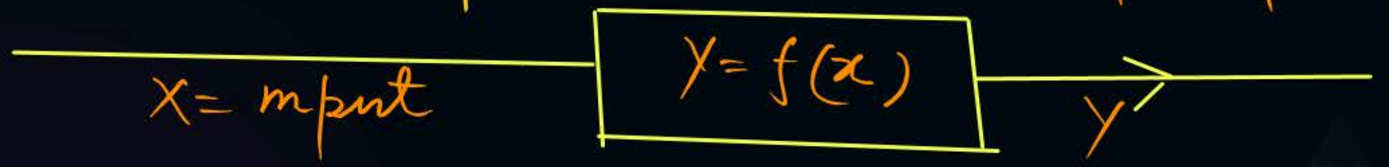
X = Independent var (Assign

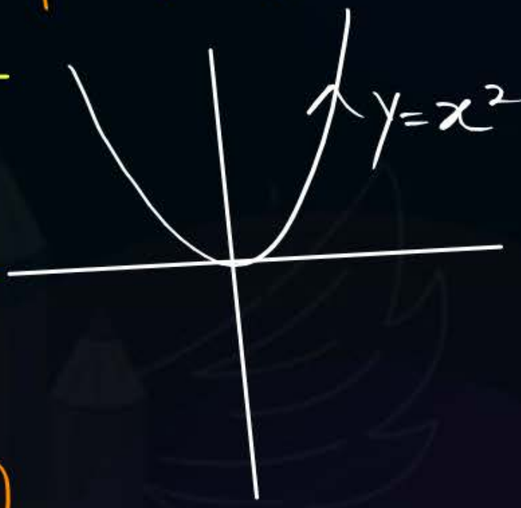
Y = obtain

Dependent var

$Y = x^2$ $\quad X \in$ Real No

X	1	2	3	4
Y	1	4	9	16





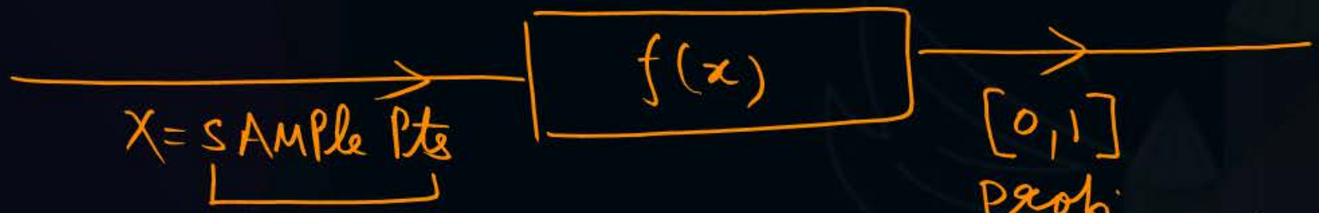


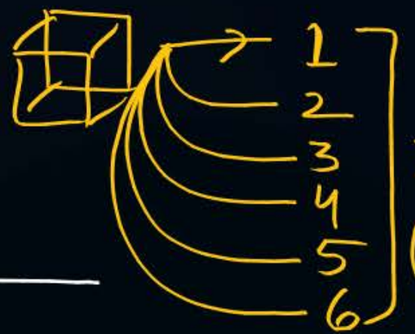
Photo – sketch / graph / curve

Throwing A Die

1)



X = No. of dots



Integer (Arrival)



Input

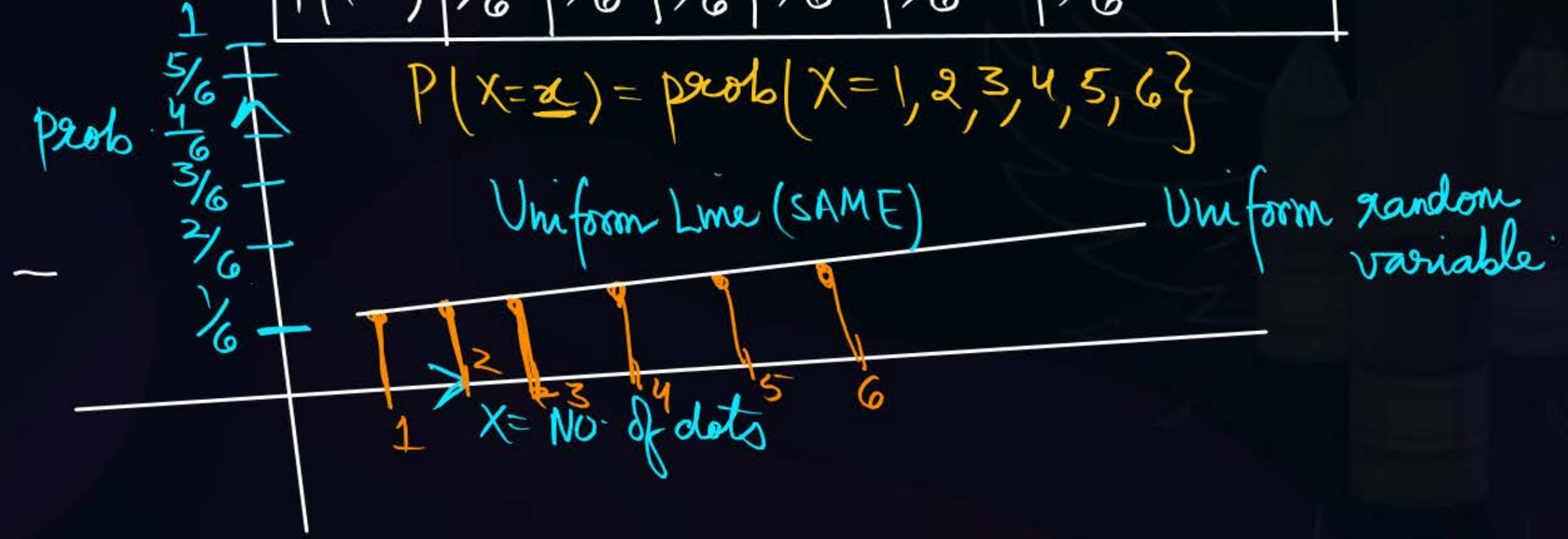
$X = 1, 2, 3, 4, 5, 6$

output prob $\left(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}\right)$

Table:

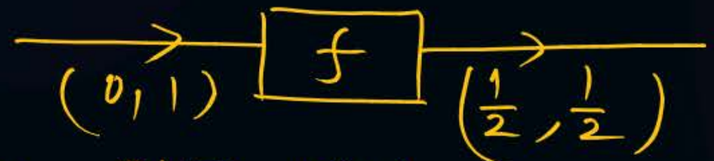
X	1	2	3	4	5	6
$P(X=x)$	1/6	1/6	1/6	1/6	1/6	1/6

$P(X=x) = \text{prob}\{X = 1, 2, 3, 4, 5, 6\}$



1) Single coin

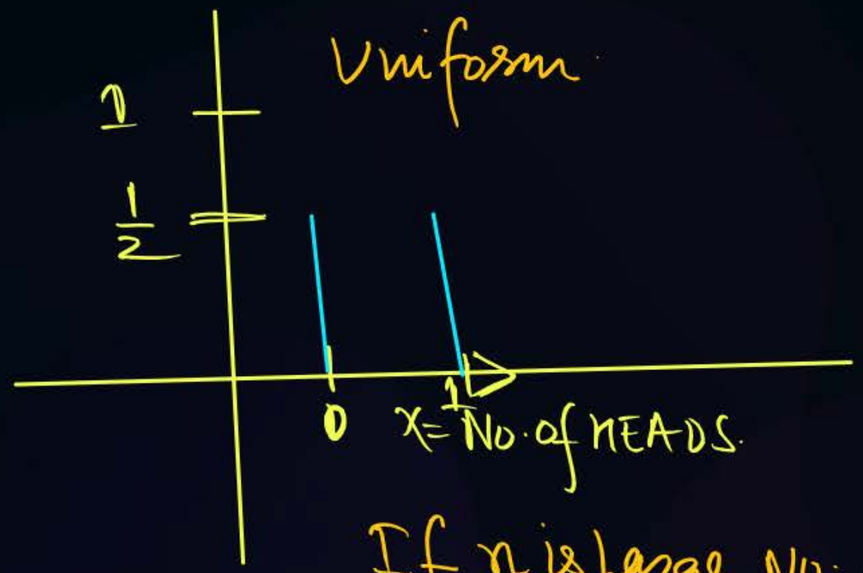
H, T



X = No. of HEAD $P(E) = \frac{n(E)}{n(S)}$

X	H (1)	T (0)
P(X=x)	1/2	1/2

$P(E) = \frac{1}{2}$



If n is large No. of trials ⟶ Gaussian random variable (bell shaped)

For Two coins

Re1, Re2

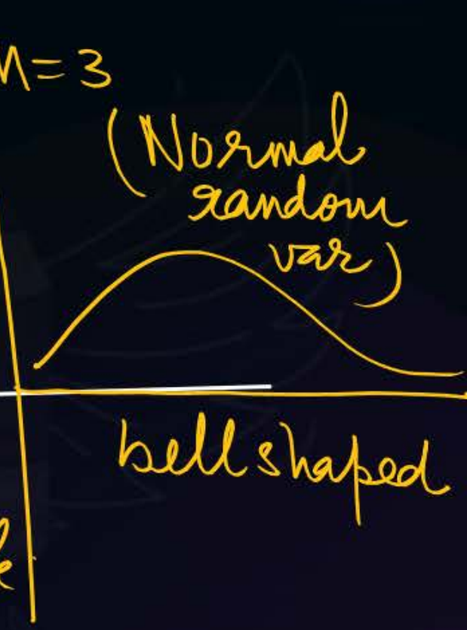
HH, HT, TH, TT

X = No. of HEADS/TAIL



X =	0	1	2
P(X=x)	1/4	2/4	1/4





- Bernoulli
- binomial
- Poisson
- Uniform (Discrete) (Arrival)

Random variable

(Discrete)

Countable
(Countable Infinite)

always defined on Integer value

- ✓ Average No of customer
- ✓ Tossing A coin
- ✓ Picking A ball
- ✓ Throwing A Die
- ✓ No of calls Per unit Time

X = random vari

$X = 0, 1, 2, 3 - -$

Continuous random

Uniform (conti)

(Waiting Time)
(Interval)
MEASURABLE

Infinite Uncountable
(Interval)

- ✓ Person Height
  $1.1 \leq X \leq 1.2$
- ✓ Discharge A Battery
  $1 \text{ min} \leq X \leq 2$
- ✓ Uniform continuous
- ✓ Normal
- ✓ Exponential (Waiting Time)

Infinite Pts – measurable

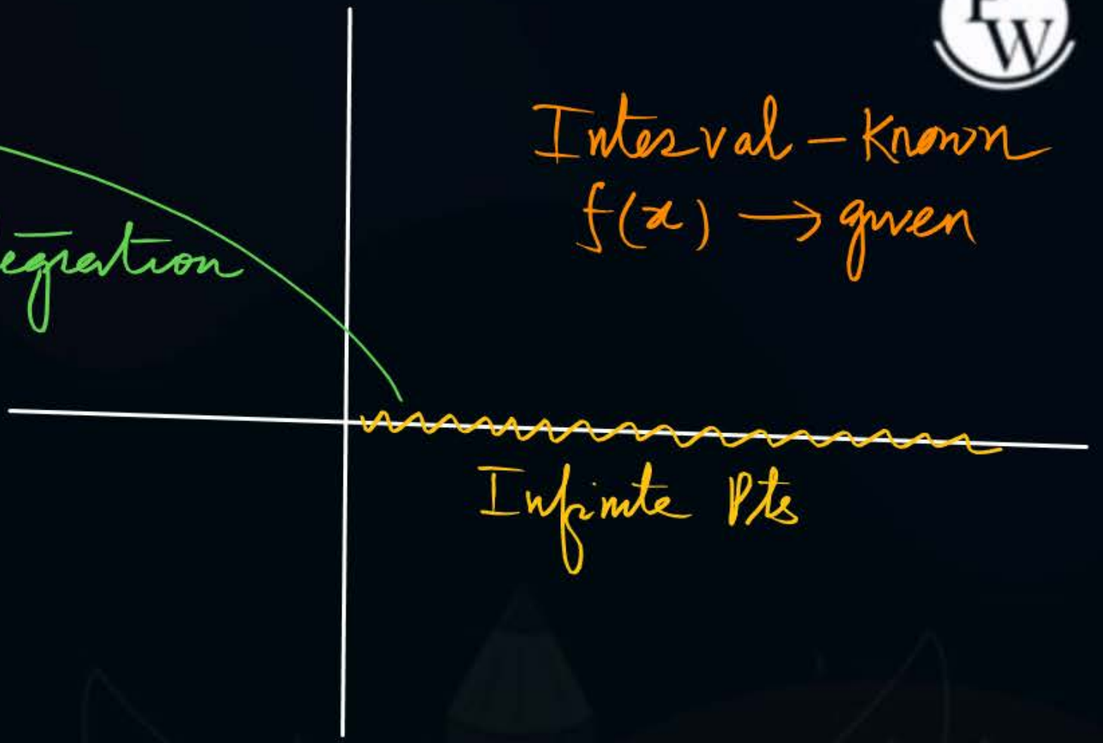
Infinite Pts Add — Tool — Integration

= Algebraic sum

$$\int_A^B f(x)\,dx$$

Interval – known

$f(x)$ → given



Distribution Function: (Discrete Random)

1) Cumulative Distribution Function:

$$F_X(x_i) = P[X \le x_i]$$

Prob. Add

$$F_X(1) = P[X \le 1] = P_1$$

$$F_X(2) = P[X \le 2] = P[X=1] + P[X=2] = P_1 + P_2$$

$$F_X(3) = P[X \le 3] = P_1 + P_2 + P_3$$

$$F_X(x_i) = P_1 + P_2 + P_3 + \cdots + P_i$$

Throwing A Die :

X = No. of dots

$X =$	1	2	3	4	5	6
$P(X=x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$F_X(1) = P[X \le 1] = P_1 = \frac{1}{6}$ ✓ > 0

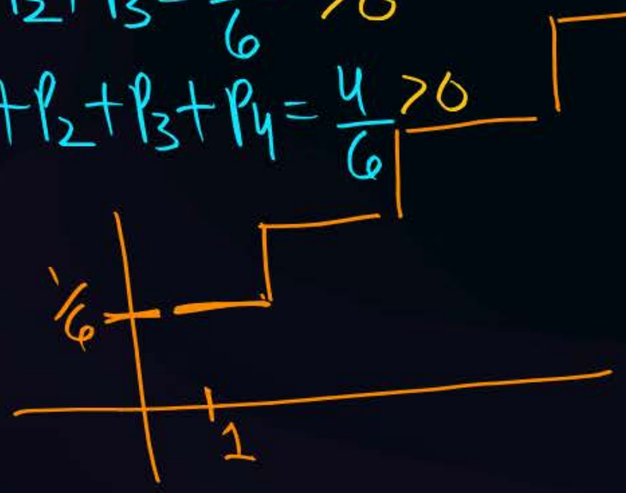
$F_X(2) = P[X \le 2] = P_1 + P_2 = \frac{2}{6} > 0$

$F_X(3) = P[X \le 3] = P_1 + P_2 + P_3 = \frac{3}{6} > 0$

$F_X(4) = P[X \le 4] = P_1 + P_2 + P_3 + P_4 = \frac{4}{6} > 0$

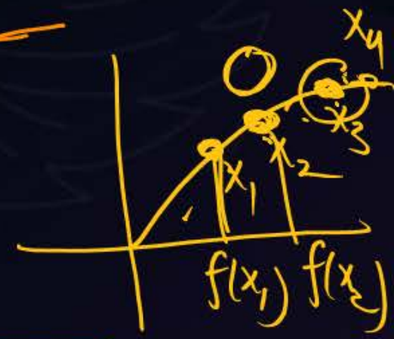
$F_X(5) = \frac{5}{6} > 0$

$F_X(6) = \frac{6}{6} = 1 > 0$









Throwing A Two Die

Die A + Die B
$X + Y$

2	3	4	5	6	7	8	9	10	11	12
$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

+	1	2	3	4	5	6
1	(2)	(3)	(4)	5	6	7
2	(3)	(4)	5	6	7	8
3	(4)	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Using Defination of cdf

$$F_X(x_i) = P[X \le x_i]$$

$F_X(2) = \frac{1}{36}$ $F_X(6) = \frac{15}{36}$ $F_X(10) = \frac{33}{36}$

$F_X(3) = \frac{3}{36}$ $F_X(7) = \frac{21}{36}$ $F_X(11) = \frac{35}{36}$

$F_X(4) = \frac{6}{36}$ $F_X(8) = \frac{26}{36}$ $F_X(12) = \frac{36}{36}$

$F_X(5) = \frac{10}{36}$ $F_X(9) = \frac{30}{36}$

✓ $F_X(-\infty) = 0$ Lower
$F_X(\infty) = 1$ Higher

✓ $F_X(x_i) \geq 0$ $P_X(x_i) \geq 0$

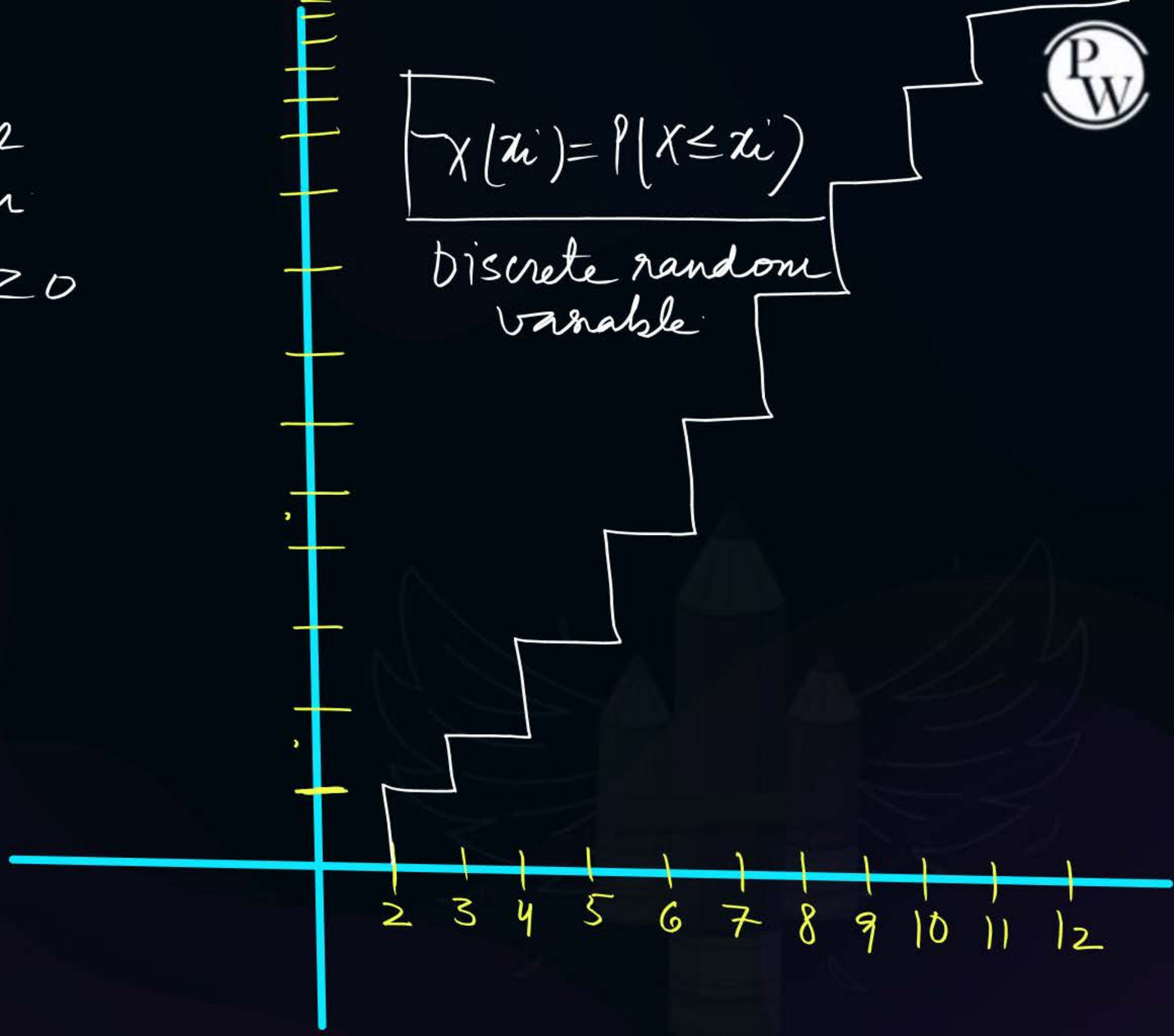
✓ Ladder Function
Staircase Function

✓ monotonically
Increasing

✓

$F_X(x_i) = P(X \leq x_i)$

Discrete random variable



Tossing A THREE coins.

✓ DPP 01 Q2

No. of HEADS = X ⟶ cdf

X	0	1	2	3
P(X=x)	1/8	3/8	3/8	1/8

$F_X(x_i) = P(X \le x_i) = P(X \le 0) = P_0 = \frac{1}{8}$

$F_X(1) = P(X \le 1) = P_0 + P_1 = \frac{1}{8} + \frac{3}{8} = \frac{4}{8}$

$F_X(2) = P(X \le 2) = P_0 + P_1 + P_2 = \frac{4}{8} + \frac{3}{8} = \frac{7}{8}$

$F_X(3) = \frac{8}{8} = 1$

Ladder





Topic : Probability and Statistics

#Q. Given that is x is a random variable in the range $[0, \infty]$ with a probability density function $\frac{e^{-\frac{x}{2}}}{k}$ the value of the constant K is.

THANK - YOU